

SYNTEEN SC46K

SOIL REINFORCEMENT GEOGRID

SC46K is a uniaxial fabric composed of high molecular weight, high tenacity multifilament polyester yarns that are woven into a stable network placed under tension. Synteen SC Series Fabrics are inert to biological degradation and are resistant to naturally encountered chemicals, alkalis and acids. Synteen SC Series Fabrics are used in a wide variety of soil reinforcement applications including reinforced soil slopes (RSS), mechanically stabilized earth (MSE) walls, and soft soil subgrade stabilization.

MECHANICAL PROPERTIES	TEST METHOD	UNIT	MINIMUM AVERAGE ROLL VALUE (MARV) ¹
Ultimate Tensile Strength (MD)	ASTM D 4595	lbs/ft (kN/m)	46,150 (673.5)
Creep Reduced Strength (MD) ²	ASTM D 4595	lbs/ft (kN/m)	17,400 (253.9)
Long Term Design Strength (MD) ³	GRI GT-7	lbs/ft (kN/m)	23,546 (343.6)

PHYSICAL PROPERTIES	TEST METHOD	UNIT	ROLL VALUE
Roll Dimensions ⁴ (Width x Length)	Measured Value	ft (m)	16.4 X 300 (5.0 X 91.44)
Roll Area	Measured Value	Yd ² (m ²)	546.66 (457.07)

¹ Minimum Average Roll Values (MARV) are calculated as the typical minus two (2) standard deviations. Statistically, it yields a 97.7% degree of confidence that any sample taken from quality assurance testing will exceed the value reported.

² Creep Reduced Strength is based on a 75-year design life at 20 °C. $RF_{CR} = 1.51$ is used based on a certified evaluation report.

³ Long-term design strength (LTDS) is calculated for a 75-year design life at 20 °C, silty sand ($D_{50} = 0.9\text{mm}$) backfill, and standard soil pH range 3 - 9. $RF_{CR} = 1.51$; $RF_{ID} = 1.18$; $RF_D = 1.1$

⁴ Custom roll lengths are available upon request.

LTDS and Reduction factors for the project specific requirements such as design life, soil type and soil pH beyond standard range 3-9 are available upon request.

SYNTEEN Technical Fabrics reserves right to amend product specifications without notice. Users / Buyers shall verify the product compliance with the project specifications or design requirements.